GREEDY METHOD

- Greedy is the most **straight forward** algorithm design technique.
- Most of the problems have **n inputs** and require us to **obtain a subset that satisfies some constraints.**
- Any subset that satisfies these constraints is called a **feasible solution**.
- We need to find a feasible solution that either **maximizes or minimizes the objective function.**
- A feasible solution that does this is called an **optimal solution**.
- The greedy method is a simple strategy of progressively building up a solution, one element at a time, by choosing the best possible element at each stage.
- At each stage, a decision is made regarding whether or not a particular input is in an optimal solution.
- This is done by considering the inputs in an order determined by some selection procedure.
- If the inclusion of the next input, into the partially constructed optimal solution will result in an infeasible solution then this input is not added to the partial solution.

"Greedy Algorithm solves problems by making the best choice that seems best at the particular moment"

Components of Greedy Algorithm

Greedy algorithms have the following five components −

1. **A candidate set** − A solution is created from this set.
2. **A selection function** − Used to choose the best candidate to be added to the solution.
3. **A feasibility function** − Used to determine whether a candidate can be used to contribute to the solution.
4. **An objective function** − Used to assign a value to a solution or a partial solution.

5. **A solution function** − Used to indicate whether a complete solution has been reached.

CONTROL ABSTRACTION

```
Algorithm Greedy (a, n)
{
// a is an array of n inputs
Solution: =Ø;
for i: =1 to n do
{
s: = select (a);
if (feasible (Solution, s)) then
{
Solution: = union (Solution, s);
}
else
reject (); // if solution is not feasible reject it.
}
return solution;
```

- The function select selects an input from 'a', removes it and assigns its value to's'.
- Feasible is a Boolean valued function, which determines if 's' can be included into the solution vector.
- The function Union combines 's' with solution and updates the objective function.

Three important activities

1. A selection of solution from the given input domain is performed, i.e. s:=select(a).
2. The feasibility of the solution is performed, by using feasible'(solution, s)'and then all feasible solutions are obtained.

3. From the set of feasible solutions, the particular solution that minimizes or maximizes the given objective function is obtained. Such a solution is called optimal solution.

PROBLEMS SOLVED USING GREEDY STRATEGY

- **The Fractional Knapsack Problem**
- **Prims' and Kruskal's Algorithms - Minimal Spanning Tree**
- **Job Sequencing Problem.**